



From Bricks to Forms



DigiPass

Know Your Products!

Building Digital Product Passports with SHACL

A visual approach to structured data

No coding background required

Know Your Products!

COORDINATION AND KNOWLEDGE SHARING ACROSS MATERIALS DEVELOPMENT COMMUNITIES (CSA):
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WIKKI LIMITED, UK participant in Horizon Europe Project DigiPass, is supported by UKRI grant number 10100819: DigiPass



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Digital Product Passports Are Hard

✗ TODAY

- Company A says: 'weight: 5kg'
- Company B says: 'mass: 5000g'
- Company C says: '5' (just a number!)

→ Computers can't understand this
→ Manual data entry & errors
→ Compliance headaches

✓ WITH SHACL

- Product.weight = 5
- Must be a number
- Must have a unit
- Unit can only be: kg, g, lb, oz

→ Machines can validate it
→ Forms guide data entry
→ Always consistent



What If Data Had Grammar?

Like grammar rules in language, data needs rules too.

✗ "The product weighs about 5kg usually" (ambiguous)

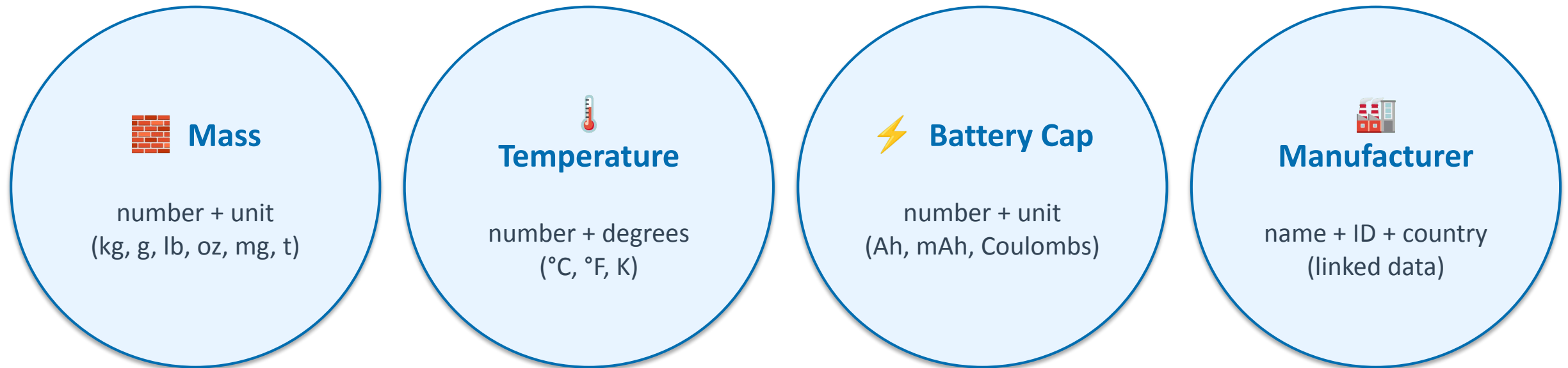
✓
Product.weight = 5
 type → number
 unit → kg (or g, lb, oz)
required → yes
checked → automatically

SHACL is the W3C standard for writing these rules.



Lego Bricks for Data Schemas

Instead of writing rules from scratch every time → create reusable bricks once, snap them together.



Define once → reuse across every product schema in your organisation.



Creating a Reusable Brick

Point-and-click — no coding needed:

Name	BatteryCapacity	
Path	ex:batteryCapacity	
Datatype	xsd:decimal	
Class ✦✦	qudt:ElectricCharge	← triggers magic

Result →

Unit dropdown appears automatically:

- ▼ Ampere-hour (Ah)
- Milliampere-hour (mAh)
- Coulomb (C)
- Kiloampere-hour (kAh)

The system looks up applicable units from the QUDT ontology — no manual list needed.



Assembling Bricks into a Product Schema

```
EV Battery Schema
├── Battery Identification
│   ├── Model Number      (text)
│   ├── Serial Number     (text)
│   └── Manufacturer      (linked data)
├── Physical Properties
│   ├── Mass              ← [Mass brick]          → unit dropdown
│   └── Capacity           ← [BatteryCapacity]     → unit dropdown
└── Environmental
    ├── Carbon Footprint  ← [CO2 brick]          → unit dropdown
    └── Recyclability      ← [Percent brick]       → % field
```

Visual tree editor — drag bricks in, set connections, done.

One Source → Many Outputs



The Magic: One Brick, Three Things

Brick (JSON)

```
{
  name: 'Mass',
  datatype: xsd:decimal,
  class: qudt:Mass,
  units: [kg,g,lb,...]
}
```

SHACL Rules

```
sh:property [
  sh:path ex:mass ;
  sh:datatype xsd:decimal ;
  sh:class qudt:Mass ;
  sh:in (kg g lb oz) ;
] .
```

User Form

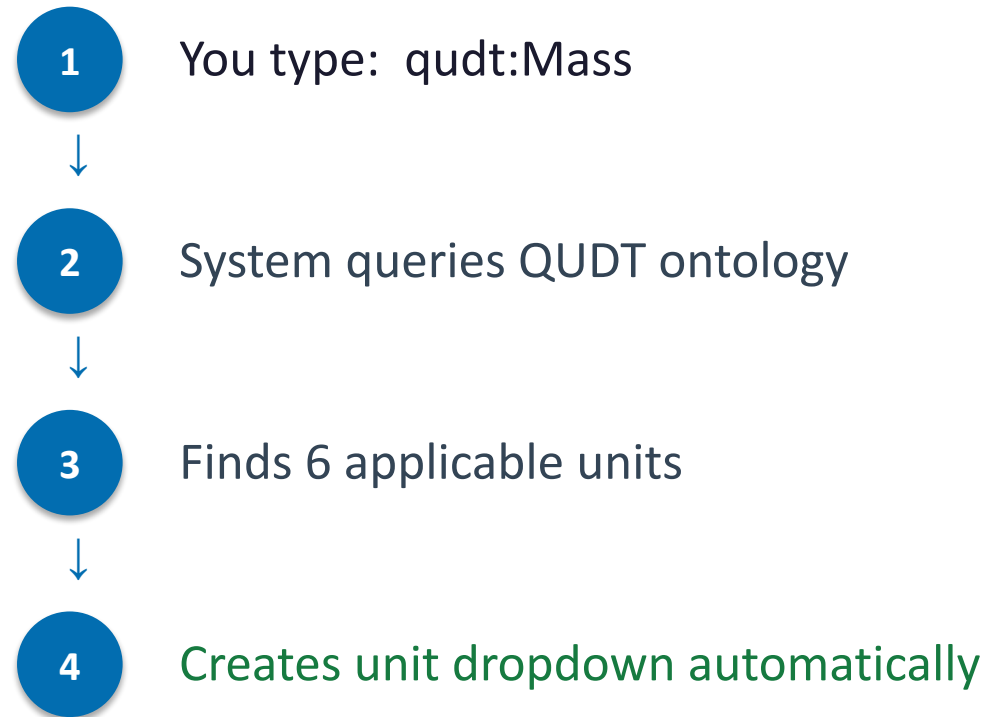
Mass

5	▼ kg
---	------

Change brick once
→ all forms update



Smart Widgets from Context



Works for any physical quantity: temperature, pressure, energy, voltage ...



How It All Connects

Shared Ontologies QUDT · Schema.org · FOAF · SKOS (17 vocabularies)

Brick Library Reusable building blocks (JSON)

Schema Composer Assemble bricks into product blueprints

SHACL Exporter Generates validation rules (Turtle)

Form Generator HTML interface the data-entry clerk sees

Validated Data Ready for Digital Product Passports



Why This Matters

	Before	After
Schema definition time	3 months	1 week
Unit configuration	manual lookup	automatic
Validation errors	50+ (found late)	2 (caught early)
Consistency	varies per team	standardised
Non-technical authors	impossible	point-and-click

Key Takaways



Bricks

Reusable, semantic data components — define once, use everywhere



SHACL

Standardised validation rules — machines can check your data



Enrichment

Ontology-driven smart widgets — unit dropdowns appear automatically



Forms

Automatic user interfaces — no coding needed for data entry



Reuse

Change a brick once → every form using it updates

Getting Started



One Download, One Command

- 1 Download** Clone or download the repository from GitHub
- 2 Run** Clone or download the repository from GitHub
- 3 Open** Clone or download the repository from GitHub
- 4 Your data** Clone or download the repository from GitHub

Questions?



Q: Can non-programmers use this?

A: Yes — brick_app and schema_app are point-and-click web tools.

Q: What about our own vocabulary?

A: Load any RDF/OWL ontology — enrichment works with it automatically.

Q: Does it connect to our database?

A: Exported SHACL validates data; can export to JSON/XML/RDF for any system.



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